Ada Sahin

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PROFILE

Nuclear Energy Engineering student specializing in feature engineering, control, and embedded systems with a focus on implementing machine learning and deep learning into engineering applications. Proficient in Python, C++ and microcontrollers (Arduino and STM32) with several experiences on building physics simulations; ANN, CNN based systems. Aiming to pursue graduate studies combining nuclear engineering with AI driven monitoring systems.

EDUCATION

Hacettepe University — Ankara, Turkey
B.Sc. in Nuclear Energy Engineering — 2025 – Present

Ankara Ataturk High School — Ankara, Turkey
Science Track, GPA: 96.71/100 — 2021 – 2025

EXPERIENCE

Ventura Yazılım — Ankara, Turkey
CAD Designer & Software Developer — 2026 – Present

Hacettepe Yapay Zeka Topluluğu — Ankara, Turkey
Active Project Developer, Machine and Deep Learning Dep — 2026 – Present

PROJECTS

Active Project Developer, Machine and Deep Learning Dept. – TEKNOFEST — 2026

- Engineered data visualization and preprocessing pipelines for large scale sensor and image datasets.
- Built and trained CNN, ANN, and RNN models using PyTorch for competition stage evaluation.

Reactor Detection and Monitoring System — 2026

Multi input alert system customized to individual reactor characteristics.

- Engineered kinetic equations of reactor physics model paired with a custom sensor data simulator.
- Designed a hybrid monitoring core combining physics modeling with ML anomaly detection.

Muon Detector — 2026

Independently designed hardware and software prototype

- Designed and built a detector to characterize muon flux distribution and forecast future measurements.
- The model's evaluation using metrics such as F1-score, accuracy, and the confusion matrix indicates consistent performance.

Health Risk Prediction Models — 2026

Collection of ML models for disease risk estimation

- Built independent ML models estimating individual and public health risk across multiple diseases from clinical datasets.
- Reached up to 0.96 accuracy on several models with no significant evidence of overfitting.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Arduino, SQL
- **Libraries & Tools:** NumPy, Pandas, SciPy, PyTorch, scikit-learn, Matplotlib, Git, GitHub
- **Domain Knowledge:** Embedded Systems, Neural Networks, Machine Learning, Deep Learning

CERTIFICATIONS & COURSES

Advanced Python Programming	BTK Academy, 2025
Machine Learning Algorithms with Python	BTK Academy, 2026
Arduino with Projects (Advanced Level)	BTK Academy, 2026
PyTorch Applications (Advanced Level)	BTK Academy, 2026
CS 230 (DEEP LEARNİNG)	STANFORD ONLINE, 2026
CS 231N (Deep Learning for Computer Vision)	STANFORD ONLINE, 2026

LANGUAGES

Turkish (Native) — English (Advanced) — Russian (Intermediate) — German (Intermediate)